

MM-DeepResearch: Reproduction

A Simple and Effective Multimodal Agentic Search Baseline
Yao et al. (ByteDance, NTU) · ICML 2026 Paper #631 · arXiv:2603.01050

Reported Benchmark Results (Paper Table 1)

SimpleVQA

62.6 +18.1

MMSearch

67.8 +30.4

LiveVQA

54.3 +11.3

FVQA

55.4 +15.3

InfoSeek

57.6 +15.8

BrowseComp-VL

80.4 +11.2

8B Average: 63.0% (+17pp over Qwen3-VL-8B) · 32B Average: 65.3% (+14.9pp over Qwen3-VL-32B)

C1

Hyper-Search QA

Hypergraph-based QA generation: builds depth-D hypergraphs from web sources, produces intra- and inter-hyperedge QA pairs → Hyper-Search-3K dataset.

Supported

 Methodology well-specified; generation code not open-sourced.

C2

DR-TTS Trajectories

4 tool types trained via RL experts → tree search recomposition → DR-TTS-10K SFT trajectories.

- text_search (E5 FAISS)
- image_search_by_text_query (Jina-CLIP)
- image_search_by_lens (image similarity)
- model_search (expert LLM)

Supported

 Tools confirmed; generation code not in repo.

C3

Training Pipeline

CONFIRMED

 — Full code audit.

- SFT base: HuanjinYao/MM-DeepResearch-8B-SFT
- GRPO RL: n=5 rollouts, lr=1e-6, 35 epochs
- Max 4 tool turns, max len 70K
- Format reward (0.1): tag structure
- Accuracy reward (0.9): EM + LLM judge
- 17 data sources
- 8x H20 GPUs

CONFIRMED

C5

Ablation: Offline Corpus

E5 (text) + Jina-CLIP (image) embeddings with FAISS GPU indexes, matching paper.

Supported

 Infrastructure confirmed.

C4

Benchmark Results

Paper claims 63.0% (8B) / 65.3% (32B) average accuracy across 6 benchmarks.

Model	Avg	Δ vs base
Qwen3-VL-8B	46.0	—
MM-DeepResearch-8B	63.0	+17.0
Qwen3-VL-32B	50.4	—
MM-DeepResearch-32B	65.3	+14.9

Not Verified

 Needs 2x A100/H100 + search API keys.

Data

Resources

Model (8B):
https://huggingface.co/HuanjinYao/MM-DeepResearch-8B
Corpus:
https://huggingface.co/datasets/HuanjinYao/MM-DeepResearch-corpus
GitHub:
https://github.com/HJYao00/MM-DeepResearch

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Conclusion

Claim	Status
C1: Hyper-Search	Supported
C2: DR-TTS	Supported
C3: Training	Confirmed
C4: Benchmarks	Paper only
C5: Ablation	Supported

Core training pipeline is fully reproducible. Data generation pipelines not open-sourced. Benchmark verification requires GPU compute + paid API keys.

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Architecture

- Base: Qwen3-VL-8B/32B-Instruct
- Training: VeRL + Ray + SGLang
- Retrieval: FAISS GPU (E5 + Jina-CLIP)
- Reward: Format + EM + LLM Judge
- Tools: text_search, image_search, lens_search, model_search

